MILITARY SPECIFICATION

TRANSPONDER SET

AN/DKW-3A(V)

1. SCOPE

1.1 Scope. This specification establishes the requirements for the AN/DKW-3A Transponder Set, hereinafter called the transponder (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-W-5088	Wiring, Aerospace Vehicle.
MIL-E-5400	Electronic Equipment, Aerospace, General Specification for.
MIL-T-5422	Testing, Environmental, Aircraft, Electronic Equipment.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Engineering Specifications and Standards Department (ESSD) Code 93, Lakehurst, New Jersey 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5821

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MIL-E-17555	Electronic and Electrical Equipment, Accessories and Repair Parts: Packaging and Packing of.
MIL-T-18303	Test Procedures; Preproduction, Acceptance and Life for Aircraft Electronic Equipment, Format for.
MIL-N-18307	Nomenclature and Identification for Electronics, Aeronautical and Aeronautical Support Equipment Including Ground Support Equipment.

STANDARDS

MILITARY

MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461	Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interface.
MIL-STD-704	Aircraft Electric Power Requirements.
MIL-STD-781	Reliability Design Qualification and Production Acceptance Tests: Exponential Distribution
MIL-STD-785	Reliability Program for System Equipment Development and Production.
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of.
MIL-STD-965	Parts Control Program.
MIL-STD-2074	Failure Classification for Reliability Testing.

FEDERAL

FED-STD-595	Colors

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified, a sample shall be subjected to first article inspection (see 4.4 and 6.3).

3.2 Parts and materials. In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing, the following shall govern:

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a. Microelectronic technology shall be considered, and microelectronic items shall be in accordance with requirements specified herein.

b. Other parts and materials requirements shall be in accordance with MIL-E-5400.

c. Nonrepairable subassemblies shall be used in accordance with MIL-E-5400. When previously produced models of this transponder did not use nonrepairable subassemblies, the design shall not be changed to employ nonrepairable assemblies without the approval of the procuring activity.

3.2.1 Nonstandard parts and materials approval. Approval for use of nonstandard parts and materials (including transistors and diodes) other than microelectronic devices shall be obtained in accordance with MIL-E-5400. Approval for microelectronic devices shall be obtained in accordance with MIL-STD-965, Procedure I, except that second source development and interchangeability testing shall not be required (see 6.2.2).

3.2.2 Microelectronic modular assemblies. When used, microelectronic modular assemblies shall be in accordance with the requirements of MIL-STD-965.

3.2.3 Modules. Conformal coatings, encapsulants, embedments or potting materials used with modular assemblies containing integrated circuits and discrete parts shall be easily removable without damage to the assembly.

3.3 Design and construction. The transponder shall be in accordance with all the applicable requirements of MIL-E-5400 for design, and construction, except as otherwise specified herein.

3.3.1 Total weight. The total weight of the transponder, excluding cables, shall be a minimum consistent with good design and shall be not greater than 4.54 kilograms (kg) (10 pounds [lbs]).

3.3.2 Reliability. The contractor shall conduct a reliability program in accordance with MIL-STD-785. On a reorder from a contractor who is producing the transponder, the program previously approved (see 6.2.2) may be continued unless otherwise specified in the contract or order (see 6.2.1).

3.3.2.1 Operational stability. The transponder shall operate as required herein, continuously or intermittently for a period of not less than 450 hours without the necessity for readjustment of any controls which are inaccessible to the operator during normal use.

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3.3.2.2 Operating life. The transponder shall have a total operating life of 5000 hours with minimal servicing and replacement of parts. The contractor shall prepare a repairable parts list for review by the procuring activity (see 6.2.2). This list shall identify parts requiring scheduled replacement.

3.3.2.3 Reliability in mean time between failure (MTBF). The transponder, including any built-in-test provisions, shall have an upper test MTBF (θ_0) of 450 hours when tested as specified in 4.5.2.

3.3.3 Cabling and connections. Cabling and connections shall be as specified in 3.3.3.1 and 3.3.3.2.

3.3.3.1 Cables and connectors. The transponder shall provide for the use of wiring and connectors in accordance with MIL-E-5400.

3.3.3.2 Interconnection cabling. The transponder shall be capable of required operation using external wiring in accordance with the requirements of MIL-W-5088. External cable shielding may be used as necessary to meet interference control requirements.

3.3.4 Interchangeability. The transponder shall be in accordance with the interchangeability requirements of MIL-E-5400.

3.3.5 Interference control. The generation of radio interference by the transponder and vulnerability of the transponder to radio interference shall be in accordance with the following requirements of MIL-STD-461, as modified herein:

a. CE03 Tne narrowband limit shall increase from 20 dBuA at 2 MHz to 100 dBuA at 100 kHz and remain at this 100 dBuA to 15 kHz. The broadband limit shall increase from 50 dBuA/Mhz at 2 MHz to 160 dBuA/MHz at 100 kHz and remain at 160 dBuA/MHz to 15 kHz.

b. CE06 (See 3.5.1.1.3 and 3.5.1.2.4).

c. CS01

d. CS02

e. CS03 The transponder performance shall not be degraded by two equal amplitude intermodulation signal sources at levels lower than -50 dBm while responding to a normal interrogation at -86 dBm $\pm$2 dB.

f. CS04 (See 3.5.1.1.2).

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g. CS06

h. RE02 The narrowband and broadband limits shall be relaxed 20 dB below 100 kHz. The transmitter fundamental shall be exempt.

i. RS02

j. RS03 The receiver fundamental shall be exempt.

3.3.6 Provisions for maintainability. The contractor shall provide for maintainability in accordance with the requirements of MIL-E-5400.

3.3.7 Nomenclature, nameplates and identification marking. Nomenclature, serial number assignment, nameplate approval and identification marking shall be in accordance with MIL-N-18307 (see 6.2.2).

3.3.8 Standard conditions. The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

a. Temperature: Room ambient (25°C ±10°C)

b. Altitude: Normal ground

c. Vibration: None

d. Humidity: Room ambient up to 95 percent relative humidity

e. Input voltage: 28.0 ±0.5 Vdc

3.3.9 Service conditions. The transponder shall operate as specified herein under any of the environmental service conditions or reasonable combination of these conditions as specified in MIL-E-5400 for Class 2 equipment, except as specified herein.

3.3.9.1 Vibration. The transponder shall operate as specified herein when subjected to the vibration requirements of Figure 2 curve IIIa of MIL-E-5400.

3.3.9.2 Temperature and temperature-altitude. Intermittent operation at +95°C is not required. Operation below -40°C is not required.

3.3.10 Warm-up time. The time required for the transponder to warm-up prior to operation shall be kept to a minimum and shall be not greater than 5 minutes under standard conditions and 15 minutes at extreme temperature conditions as identified in MIL-E-5400 for Class 2 equipment and modified by 3.3.9.2. herein.

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3.3.11 Input electrical power. Input electrical power shall be as specified in 3.3.11.1 and 3.3.11.2.

3.3.11.1 Operating power. The transponder shall meet all applicable requirements of MIL-STD-704 and shall give specified performance when energized from the following power sources having characteristics and limits as defined in MIL-STD-704, except as specified herein. The power required shall be not greater than the amounts herein:

a. DC power, 28 Vdc - 30 watts normal telemetry operations at 9 percent duty factor, 55 watts when transmitting continuously. Reverse polarity protection on the +28 Vdc inputs shall be provided.

b. DC power, 5 Vdc - .025 watts telemetry reference.

3.3.11.2 Degraded performance. Degraded performance is permitted for voltage transients with durations of 50 microseconds (μs) to 0.5 second during normal electric system operation. Operation shall return to normal within one second with no resulting damage to the transponder. Degraded performance is also permitted for undervoltage conditions of +22.0 to +24.0 Vdc. Transmitter power output may decrease up to 2 dB and receiver sensitivity may decrease up to 3 dB under this reduced voltage condition.

3.3.12 Form factor. The transponder shall be in accordance with the outline drawing of FIGURE 1.

3.4 Performance. Unless otherwise specified herein, values set forth to establish requirements of required performance apply to performance under both standard and extreme environmental conditions as identified in MIL-E-5400 for Class 2 equipment and modified by 3.3.9.2 herein.

3.4.1 Operation. The transponder shall not be provided with a power switch and is intended to be activated when power is applied to the host vehicle. Only setting of TARGET ADDRESS (two multi-position rotary switches) shall be required prior to use.

3.4.2 Frequency characteristics. The transponder shall be capable of receiving and transmitting on any of the 160 Integrated Target Control System (ITCS) channels between 4.4 and 4.8 GHz, when equipped with the appropriate crystals and properly tuned. Separation between transmit and receive frequencies shall be not less than 195 MHz.

3.4.3 Uplink modulation characteristics. The uplink modulation characteristics shall be as specified in 3.4.3.1 and 3.4.3.2.

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3.4.3.1 Uplink modulation type. The transponder shall be capable of receiving and processing standard ITCS uplink signals, which are frequency modulated (FM) with a peak deviation of ±200 KHz ±5 percent.

3.4.3.2 Uplink modulation format. The baseband information shall consist of digital command data linearly summed with a data-coherent square wave ranging tone. The data shall be binary in level with zero average value. The data shall be at a sub-bit rate of 51.30 kilobits per second (kbps) and shall be exactly 0.50 of the 102.60 kHz range tone frequency. The positive going transitions of the 102.60 kHz square wave shall lead the data bit transitions by 90 degrees. The data shall produce a carrier deviation of ±100 kHz ±5 percent. A positive level, as specified herein, shall produce a negative carrier frequency deviation.

3.4.4 Downlink modulation characteristics. The downlink modulation characteristics shall be as specified in 3.4.4.1 through 3.4.4.3.

3.4.4.1 Downlink modulation type. The transponder shall transmit the standard ITCS downlink signal which is phase modulated with a peak phase deviation of 1.4 radians ±10 percent.

3.4.4.2 Downlink modulation format. The transponder shall operate in a carrier only mode or telemetry mode (TM), as determined by the command message. In switching from carrier only to TM, data transmission on the carrier frequency shall not shift more than ±5 kHz. Telemetry data consists of a 32 bit preamble followed by 17 words in which each word contains 16 bits. Of the 16 bits, 15 are actual data and the 16th is an odd parity bit.

3.4.4.2.1 Parallel mode. The basic, non-varying portion of the ITCS downlink telemetry format for the parallel mode shall be as shown in FIGURE 2.

3.4.4.2.2 Serial mode. The basic, non-varying portion of the ITCS downlink telemetry format for the serial mode shall be as shown in FIGURE 3.

3.4.4.3 Telemetry mode. In the telemetry mode the baseband information shall consist of digital data which bi-phase modulates a 102.60 kHz sinusoidal subcarrier that is the regenerated ranging tone. The data rate shall be 0.50 of the subcarrier frequency (51.30 kbps) and shall be coherent. A data bit "one" is specified as two cycles of subcarrier whose phase is unaltered (±2.0 degrees) while a data bit "zero" is specified as two cycles of subcarrier whose phase has been shifted 180 degrees (±2.0 degrees). Data bit transitions shall coincide with zero crossings of the ranging tone. The subcarrier shall phase modulate the carrier in such a manner that the positive half-cycles of the subcarrier cause a decrease in phase shift from that of the unmodulated carrier while the negative half-cycles of the subcarrier cause an increase in phase shift in the carrier.

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3.4.5 Command characteristics. Command characteristics shall be as specified in 3.4.5.1 through 3.4.5.4.

3.4.5.1 Command message format. The basic, non-varying portion of the ITCS command message format shall be as shown in FIGURES 4 and 5 for the parallel and serial modes, respectively. Each command data bit of words 1 through 17 shown in the figures shall correspond to two 51.30 kbps sub-bits, due to Manchester coding. The data bit rate thus shall be 25.65 kbps. A Manchester coded data bit "one" shall be a sub-bit "zero" (negative level) followed by a sub-bit "one" (positive level), and a data "zero" shall be the opposite.

3.4.5.2 Preamble. The preamble shall consist of six identical 7 bit segments (14 sub-bits). Each of the 7 bit segments shall be encoded into 14 subbits in the following sequence: 10101010111000. The unique sequence of the last six sub-bits shall not appear elsewhere in the command message. The preamble shall be followed by 17 words of ten bits each, and each ending in an odd parity bit.

3.4.5.3 Postamble. A command message postamble shall be transmitted after the last word (word 17) to facilitate antenna selection based on received signal strength. The postamble shall consist of a series of Manchester coded "zero" data bits transmitted for a period of time equal to the length of two command words (780 μs).

3.4.5.4 Command output characteristics. The 40 parallel command outputs and the serial outputs (data, sync, and clock) of the transponder shall be digital outputs, representing either discrete or proportional commands or synchronizing functions. A command output "zero" is specified as +0.25 ±0.25 Vdc, and shall not be required to sink more than 2 mA of current. A command output "one" is specified as +3.7 ±1.3 Vdc, and shall not be required to source more than 0.5 mA of current. Each command output, its pin assignment and format location shall be in accordance with TABLE I. All parallel command outputs shall initiate to the zero state upon initial application of power to the transponder and shall remain in the zero state until changed by a valid command message.

3.4.6 Telemetry input characteristics. The telemetry characteristics shall be as specified in 3.4.6.1 and 3.4.6.2.

3.4.6.1 Parallel mode telemetry input characteristics. The telemetry inputs shall consist of 10 discrete (on/off) and 10 proportional inputs. The telemetry inputs with their respective pin assignments and format locations shall be as shown in TABLE I.

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3.4.6.1.1 Discrete TM inputs. A discrete telemetry input "zero" is specified as a voltage level less than +1.0 Vdc. In addition, a "zero" shall be transmitted if the input is negative, grounded or an open circuit. A discrete telemetry input "one" shall be transmitted for an input voltage range from +2.4 to +100 Vdc. Nominal input impedance shall be 50K ohms.

3.4.6.1.2 Proportional TM inputs. A proportional telemetry input is specified as a dc analog voltage whose nominal range is 0 to +5 Vdc. Inputs from -3 to +8 Vdc shall be accepted without affecting normal operation. Nominal input impedance of each channel shall be 1 megohm. The analog-to-digital conversion of each channel shall be either 10 bits or 5 bits (see TABLE I).

3.4.6.2 Serial mode telemetry input characteristics. The serial telemetry data input shall be the non-inverting input of a comparator that is pulled up to the transponder +5 Vdc through 22K ohm resistor. A discrete telemetry input "zero" is specified as a voltage level less than +1.0 Vdc. A discrete telemetry input "one" shall be transmitted for an input voltage range from +1.5 to +5.5 Vdc or an input open. Nominal input impedance shall be 22K ohms to the +5 Vdc bus. The serial mode telemetry inputs with their respective pin assignments and format locations shall be as shown in TABLE I, J5 connector.

3.5 Detail requirements. Detail requirements shall be as specified in 3.5.1 through 3.5.3.

3.5.1 Contents. The transponder shall contain the circuit groups specified in 3.5.1.1 through 3.5.1.5.

3.5.1.1 Receiver. The receiver shall be of the superheterodyne type and capable of being tuned to any one of the 160 ITCS channels when equipped with the appropriate crystal. The specific frequency shall be specified in the contract or purchase order (see 6.2.1).

3.5.1.1.1 Sensitivity. When an rf signal modulated in accordance with 3.4.3 at a level between -90 dBm and 0 dBm is applied to the transponder antenna connector, the probability of receiving and decoding a correct command function shall be at least 0.77, and the command error probability shall be less than 0.82×10^{-8}. The probability of transmitting a telemetry message shall be 0.90 and is based on the correct decoding of word 1 in the received command message.

3.5.1.1.2 Selectivity. The receiver shall provide at least 60 dB of rejection to signals beyond $\pm$11.5 MHz of its tuned frequency and at least 80 dB of rejection to signals beyond $\pm$30 MHz of its tuned frequency; however, rejection of image frequencies need not be greater than 60 dB.

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TABLE I. Transponder connector pin assignments.

Pin	Function	Pin	Function
J3-	Power connector	J4-	Telemetry connector
A	+28 Vdc input power	h	TM start test point
B	+28 Vdc input power	j	TM data test point
C	+28 Vdc return and ground	J5-	Command connector
D	+28 Vdc return and ground	A	W3, B1, digital output
E	+22 Vdc test point	B	W3, B2, digital output
F	+15 Vdc test point	C	W3, B3, digital output
G	+5 Vdtest point	D	W3, B4, digital output
H	-15 Vdc test point	E	W3, B5, digital output
J	Spare	F	W3, B6, digital output
K	Spare	G	W3, B7, digital output
		H	W3, B8, digital output
J4-	Telemetry connector	J	Command serial data output
A	+5 Vdc reference voltage input	K	Ground
B	Ground	L	W4, B1, digital output
C	W4, B1-10, 0-5 Vdc input	M	W4, B2, digital output
D	W4, B11-15, 0-5 Vdc input	N	W4, B3, digital output
E	W5, B1-10, 0-5 Vdc input	P	W4, B4, digital output
F	W5, B11-15, 0-5 Vdc input	R	W4, B5, digital output
G	W6, B1-10, 0-5 Vdc input	S	W4, B6, digital output
H	W6, B11-15, 0-5 Vdc input	T	W4, B7, digital output
J	W7, B1-10, 0-5 Vdc input	U	W4, B8, digital output
K	W8, B1-10, 0-5 Vdc input	V	Command sync output
L	W9, B1-10, 0-5 Vdc input	W	Ground
M	W10, B1-10, 0-5 Vdc input	X	W6, B1, digital output
N	W2, B1, discrete input	Y	W6, B2, digital output
P	W2, B2, discrete input	Z	W6, B3, digital output
R	W2, B3, discrete input	a	W6, B4, digital output
S	W2, B4, discrete input	b	W6, B5, digital output
T	W2, B5, discrete input	c	W6, B6, digital output
U	W2, B6, discrete input	d	W6, B7, digital output
V	W2, B7, discrete input	e	W6, B8, digital output
W	W2, B8, discrete input	f	51.3 kHz clock
X	W2, B9, discrete input	g	Ground
Y	W2, B10, discrete input	h	W9, B1, digital output
Z	Ground	i	W9, B2, digital output
a	+5 Vdc reference output	j	W9, B3, digital output
b	Spare	k	W9, B4, digital output
c	Spare	m	W9, B5, digital output
d	Spare	n	W9, B6, digital output
e	Spare	p	W9, B7, digital output
f	Modulation test point	q	W9, B8, digital output
g	Receiver video test point	r	W10, B1, digital output

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TABLE I. Transponder connector pin assignments. - Continued

Pin	Function	Pin	Function
J5-	Command connector	J5-	Command connector
s	W10, B2, digital output	AA	Ground
t	W10, B3, digital output	BB	TM serial data input
u	W10, B4, digital output	CC	Serial mode control
v	W10, B5, digital output	DD	Failure alarm
w	W10, B6, digital output	EE	TM sync
x	W10, B7, digital output	FF	Spare
y	W10, B8, digital output	GG	Ø1 command clock
z	Ground	HH	Spare

3.5.1.1.3 Spurious emissions. Transponder generated spurious emissions shall not be greater than -50 dBm measured at either antenna connector with the transmitter inhibited.

3.5.1.1.4 Source impedance. The receiver shall provide the performance specified herein when operating from an rf source impedance of 50 ohms nominal.

3.5.1.2 Transmitter. The transmitter shall be capable of operating on any one of the 160 ITCS channels, when equipped with appropriate crystal. The specific frequency shall be as specified in the contract or purchase order (see 6.2.1).

3.5.1.2.1 Power output. The transmitter shall produce a peak power output of no less than one watt at either antenna port, and at any duty factor up to 100 percent.

3.5.1.2.2 Load impedance. The nominal transmitter load impedance shall be 50 ohms, and the transmitter shall produce its specified power output into a load VSWR of not greater than 1.5, any phase angle.

3.5.1.2.3 Modulation provisions. The transmitter shall have provisions for being phase modulated by the ITCS TM downlink format described in 3.4.4.2.

3.5.1.2.4 Spurious outputs. Spurious and harmonic outputs at either antenna port shall comply with CE06 (key down) requirements, except when the transmitter is in standby, output emissions shall not exceed -50 dBm.

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3.5.1.2.5 Spurious modulation. Spurious amplitude modulation between 10 kHz and 250 kHz shall be not greater than 5 percent. Spurious phase modulation between 10 kHz and 250 kHz shall be not greater than 0.1 radian peak. When commanded by the uplink signal to transmit an unmodulated carrier, carrier modulation due to subcarrier leakage shall be not less than 30 dB below carrier level.

3.5.1.2.6 Frequency stability. Transmitter frequency uncertainty shall be not greater than ±30 parts per million under all conditions of operation.

3.5.1.3 Antenna switch. Circuitry shall be provided to select automatically either of two antennas. The better of the two antennas shall automatically be selected whenever the signal is below -50 dBm. The antenna selection decision shall be based on received signal strength. When a signal applied to one antenna port is 2 dB stronger than the signal applied to the other port, the probability of selecting the strongest signal port shall be not less than 90 percent. If no commands are received and decoded for 0.5 ±0.1 seconds, the transponder shall alternately switch between the two antennas at a 1 cycle per second rate until reception of commands has been reestablished.

3.5.1.4 Command decoder. The command decoder shall accept the baseband demodulated signal from the receiver and process it as described in 3.5.1.4.1 through 3.5.1.4.12. A Manchester coding check shall be performed on each data bit and a parity check shall be performed on each word. If either does not check as being correct, the particular word in which the error was found shall be rejected. If the error occurs in word 1, the entire message shall be rejected and no telemetry message shall be transmitted. If the address in word 1 is correct and the word is accepted, but if the error occurs in word 2, a telemetry message shall be transmitted, but the remaining parallel output command words shall be rejected. For the parallel output commands, if the error occurs in any word from word 3 through word 10, only that particular word shall be rejected and not be decoded into the paralled command outputs. For serial command outputs, bit 10 of every word shall be a logic zero if the address of word 2 is incorrect or if the control request, word 2 bit 6 of the command message, is a logic zero. If the address of word 2 is correct and the control request bit is a logic one, then bit 10 of words 2 through 17 shall be a logic one, but only on those words that pass the bit Manchester coding and word parity checks of the transponder command decoder.

3.5.1.4.1 Decoder synchronization. Basic decoder synchronization shall be accomplished by phase locking to the 102.60 kHz ranging tone on the received uplink message. Bit and frame synchronization shall be derived from the bit pattern provided in the uplink preamble, with all clocks derived from the phase-locked 102.60 kHz tone.

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3.5.1.4.2 Decoder address capability. The first five bits of word 1 of the command format are identical to the first five bits of word 2 and shall be used to identify the target for which the message is intended. Each decoder shall have the capability to set in any of 32 possible addresses by two external controls (see TABLE II). Assuming no bit code or word parity errors are present, matching of the address in word 1 to the preset address shall be sufficient to cause a telemetry reply, but the address in both word 1 and 2 must match to allow acceptance of the commands as described in 3.5.1.4.3 through 3.5.1.4.12.

3.5.1.4.3 Telemetry mode (W1, B7). This bit shall control the type of telemetry transmission. Normal telemetry transmission shall occur when in the "zero" state. When in the "one" state, the transponder shall transmit an unmodulated carrier during the normal telemetry frame time.

3.5.1.4.4 Relay mode (W1, B8). Normal transponder operation shall occur when this bit is in the "zero" state. Then in the "one" state, telemetry data transmission shall not occur (TM inhibit).

3.5.1.4.5 Control override (W1, B9). This bit is not functional in the transponder.

3.5.1.4.6 Control request (W2, b6). When this bit is in the "zero" state, the digital parallel command outputs shall not respond to the message. When in the "one" state, the command decoder shall recognize that a message is requesting control of the command outputs. Under either condition, normal telemetry shall be transmitted. If the controlling station ceases to send the control request bit, the command parallel outputs shall no longer respond to that station's message and bit 10 of words 2 through 17 of the digital serial command output message shall be in the "zero" state. Control (from any requesting station) shall then be reestablished upon the next message received with a correct address and a control request bit. Should no control request be received for a period of 0.5 $\pm$0.1 second, the command loss circuit shall be activated and shall cause W3, B15 of the next telemetry message to be a "one". When the control request bit is in the "one" state, status of bit 10 of the serial command output words 2 through 17 shall be in the "one" state if the particular word has met Manchester Coding and odd parity checks (good word); otherwise, bit 10 shall be in the "zero" state (bad word).

3.5.1.4.7 Station identification (ID) (W2, B7-B9). Station ID is not applicable to the transponder.

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TABLE II. Target address selector switch positions.

Selector		Command message bit location				
Alphameric switch position	Numeric switch position	B1	B2	B3	B4	B5
A	0	0	0	0	0	0
A	1	0	0	1	0	0
A	2	0	1	0	0	0
A	3	0	1	1	0	0
A	4	1	0	0	0	0
A	5	1	0	1	0	0
A	6	1	1	0	0	0
A	7	1	1	1	0	0
B	0	0	0	0	0	1
B	1	0	0	1	0	1
B	2	0	1	0	0	1
B	3	0	1	1	0	1
B	4	1	0	0	0	1
B	5	1	0	1	0	1
B	6	1	1	0	0	1
B	7	1	1	1	0	1
C	0	0	0	0	1	0
C	1	0	0	1	1	0
C	2	0	1	0	1	0
C	3	0	1	1	1	0
C	4	1	0	0	1	0
C	5	1	0	1	1	0
C	6	1	1	0	1	0
C	7	1	1	1	1	0
D	0	0	0	0	1	1
D	1	0	0	1	1	1
D	2	0	1	0	1	1
D	3	0	1	1	1	1
D	4	1	0	0	1	1
D	5	1	0	1	1	1
D	6	1	1	0	1	1
D	7	1	1	1	1	1

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3.5.1.4.8 Frequency select (W6, B9). Frequency select is not applicable to the transponder.

3.5.1.4.9 Parallel command outputs. The first 8 bits of words 3, 4, 6, 9 and 10 shall be used to provide the parallel command outputs listed in TABLE I. The parallel command outputs are subject to the specified conditions of parity and coding checks, address, and control request status. Each of the parallel outputs shall remain continuously in the most recent state until updated by another accepted command. An output logic level "one" shall correspond to a command data bit "one."

3.5.1.4.9.1 Spare words. No parallel command functions have been assigned to words 5, 7, 8, and 11 through 17, and they need not be decoded into parallel outputs.

3.5.1.4.10 Serial command outputs. The transponder shall provide, in addition to the command parallel outputs, command serial data outputs consisting of the command data channel, the clock, and a synchronizing signal.

3.5.1.4.10.1 Command data channel. The command data channel shall provide the command data present on command words 2 through 17. The data shall be a 25.65 kHz digital data stream with a TTL output level. Bit 10 of each word shall be a good word/bad word indicator (see 3.5.1.4.6).

3.5.1.4.10.2 Clock. The basic 26.65 kHz clock shall be divided into four parts called phase 1, phase 2, phase 3 and phase 4. The command data serial clock shall be the phase 1 clock. Phase 1 clock shall be a positive pulse, 0.25 bit wide and whose low to high leading edge shall determine the start of each bit of the transponder command message decoder.

3.5.1.4.10.3 Sync signal. The command serial sync signal shall consist of a logic pulse whose low to high leading edge is time coincident (±200 nanoseconds [ns]) with phase 4 of bit 2 word 2 of the transponder command word and whose trailing edge is time coincident (±200 ns) with phase 4 of bit 2 word 18. Phase 4 clock shall be a positive pulse, 0.25 bit wide that determines the last 0.25 time segment of each bit of the transponder command message decoder. This sync signal shall be provided only if the transponder has received a command message and word 1 meets the following conditions:

a. The parity and Manchester coding checks on word 1 are correct.

b. The address of word 1 is correct.

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3.5.1.4.11 Command loss and failure alarm. The command loss function shall be activated when the command decoder has not detected command messages, or the command messages do not contain a control request for 0.5 ±0.1 seconds. Should these conditions persist for a longer preselectable period of time, failure alarm signal at the command output connector (J5 pin DD) shall be set to a logic "one" as specified in 3.4.5.4. The internally selectable delay periods shall be 1, 2, 4, 8 or 16 seconds (each ±20 percent), as specified in the contract or purchase order (see 6.2.1). These functions, including delay timer, shall reset upon restoration of command messages with control request.

3.5.1.4.12 Antenna hold (W1, B6). Normal transponder operation shall occur when this bit is in the "zero" state. When in the "one" state, the antenna port which is currently active shall be held until this bit is returned to a "zero", or until 0.5 ±0.1 second has expired without receiving and decoding a command message from a controlling station.

3.5.1.5 Telemetry encoder. The telemetry encoder shall accept proportional and discrete signals, from the telemetry input connector and from the command decoder; and serial digital data from the command connector J5 and process them as described in 3.5.1.5.1 through 3.5.1.5.6. The baseband information transmitted in the telemetry mode shall consist of 51.30 kbps data bi-phase modulating a phase coherent 102.60 kHz subcarrier derived from the received ranging tone. A telemetry "one" shall cause transmission of two cycles of in-phase (0°) subcarrier and a "zero" shall cause transmission of two cycles of out-of-phase (180°) subcarrier. The telemetry subcarrier and bit clock shall be synchronized to the decoder by means of the received 102.60 kHz ranging tone and the "TM start" signal. The basic ITCS downlink TM format is specified in 3.4.4.2. The telemetry format, which begins immediately following word 1 of the command format, consists of a 32-bit preamble of alternate "ones" and "zeros", starting with a "one", and followed by seventeen 16-bit words. The first six bits of word 1 shall be the "sync" sequence 100001. Bits 7 through 11 shall contain the target address code in the same order as in words 1 and 2 of the command format, as determined by the positions of the two TARGET ADDRESS switches. The sixteenth bit of each word shall be an odd parity bit.

3.5.1.5.1 Both modes. The TM specified in 3.5.1.5.1.1 through 3.5.1.5.1.5 shall be the same in either mode.

3.5.1.5.1.1 Command parity error (W1, B12). This bit shall be a "zero" if all the received and decoded commands of the previous frame from the controlling station had no parity nor Manchester coding errors. A "one" shall indicate that one or more command words were rejected due to a parity or a Manchester coding error.

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3.5.1.5.1.2 Antenna selected (W1, B13). This bit shall indicate which of the two antenna ports is operating for the current frame. A "one" shall indicate antenna port J1.

3.5.1.5.1.3 Scoring verify (W3, B1). This bit shall be a "zero".

3.5.1.5.1.4 Control verify (W3, B11). The state of this bit shall be the same as the current frame's control request bit (W2, B6) in the command uplink if the address, coding and parity are correct in command word 2; otherwise it shall be a "zero".

3.5.1.5.1.5 Command loss (W3, B15). A "one" shall indicate that no updated commands have been received for the period of 0.5 second (see 3.5.1.4.6).

3.5.1.5.2 Parallel mode. Paragraphs 3.5.1.5.2.1 through 3.5.1.5.2.7 refer to parallel mode only.

3.5.1.5.2.1 Word 1, Bits 14, 15. These bits shall be transmitted as "zeros".

3.5.1.5.2.2 Discrete inputs. The discrete telemetry inputs are specified in 3.4.6 and shall be transmitted in word 2, bits 1 through 10 as shown in FIGURE 2.

3.5.1.5.2.3 Word 2, bits 11-15. Bits 11, 12 and 13 shall be transmitted "one, zero, zero" respectively (data code for RPM). Bit 14 shall be a "one" (radar altitude range 0-5,000 feet) and bit 15 shall be a "zero" (barometric altitude range 0-40,000 feet).

3.5.1.5.2.4 Word 3, bits 2-10. These bits shall be transmitted as all "zeros".

3.5.1.5.2.5 Word 3, bits 12-14. These bits shall be transmitted as all "zeros".

3.5.1.5.2.6 Parallel mode telemetry inputs. In the parallel input mode, ten analog telemetry inputs, whose nominal range is 0 to +5 Vdc, shall be accepted through the telemetry input connector (see TABLE I). The straight binary equivalents of these signals shall be transmitted each frame, most significant bit (MSB) first in accordance with the downlink telemetry data format (see 3.4.4.2) in words 4 through 10. An analog-to-digital (A/D) converter shall convert these telemetry inputs to natural binary digital words, the lengths of which are determined by the telemetry format locations of the applicable telemetry functions. Ten bit conversion is the maximum required, and in those functions requiring 5 bits of resolution, only the most significant bits shall be transmitted.

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3.5.1.5.2.7 TM input voltage range. The nominal voltage range shall be from 0 to +5 Vdc (reference voltage input, see 3.5.1.5). Normal operation shall be that all "ones" are transmitted when the telemetry input voltage is equal to or greater than the reference voltage input, and all "zeros" shall be transmitted when the input voltage is equal to or less than zero volts. The accuracy of a 10 bit conversion shall be as follows: the input voltage at which the digital output just becomes all "zeros" shall be 0 ±0.020 Vdc. The input voltage at which the digital output just becomes all "ones" shall be within 0.025 volts of the reference voltage input. Linearity shall be within two least significant bits (LSB). Accuracy of the five-bit conversion shall be ±1 LSB.

3.5.1.5.3 Serial mode telemetry input. The serial telemetry function shall consist of two inputs and two outputs. The transponder shall provide a telemetry data bit clock and a synchronizing signal to an external telemetry encoder. The transponder shall be capable of accepting serial telemetry data and shall have an input control line that is capable of selecting the parallel data input mode or the serial data input mode. The serial telemetry functions are described in 3.5.1.5.3.1 through 3.5.1.5.4.

3.5.1.5.3.1 Serial data. The serial data input characteristics are described in 3.4.6.2. The data present at the transponder serial input shall be clocked into the transponder telemetry data register on the high to low edge of the 51.3 kHz telemetry clock. When the transponder is in the serial data mode, valid serial data will be limited to the following telemetry message bit/word times: Bits 14 and 15 of word 1; all of word 2 except bit 16; all of word 3 except bits 1, 11, 15 and 16; and all of words 4 through 17 except bit 16. Bit 16, the parity bit of all words, shall be supplied by the transponder encoder.

3.5.1.5.3.2 Sync signal. The serial synchronizing signal, TM Sync, shall consist of a logic pulse whose low to high leading edge shall occur 0.50 bit before bit 1 of word 1 of the transponder telemetry message word clock and the high to low trailing edge shall occur 0.50 bit before the end of bit 16 of word 17. This output shall be buffered to the TTL logic level and shall be capable of sourcing 0.5 mA maximum and sinking 20 mA maximum.

3.5.1.5.3.3 Telemetry data bit clock. The telemetry data bit clock shall be the transponder 51.3 kHz telemetry clock buffered to the TTL output level and capable of sourcing 0.5 mA maximum and sinking 20 mA maximum.

3.5.1.5.4 Mode control. The input on J5 pin CC shall be used to control the telemetry input mode of the transponder. When the input is open circuited, the transponder telemetry encoder shall operate in the parallel

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input mode. When the input is grounded, the transponder telemetry encoder shall operate in the serial input mode. The input characteristics shall be the same as serial telemetry data input (3.4.6.2).

3.5.1.5.5 Telemetry turnaround delay. The time delay within the transponder between the 102.6 kHz range tone modulation on the uplink rf input signals and the re-transmitted range tone as measured at the modulation input of the transmitter (J4 pin t) shall correspond to a phase shift at 102.6 kHz of 173° ±10° under all conditions of operation. Under standard conditions, the delay shall correspond to 173° ±5°. The time delay between the leading edge of uplink command word 2, bit 1, and the first bit of the downlink telemetry preamble shall be 31.5 ±1.5 µs with less than 2.0 µs rms jitter.

3.5.1.5.6 Telemetry reference voltages. The transponder reference voltage output shall be +5.0 ±0.03 Vdc and shall be capable of supplying up to 100 mA of current. The reference voltage input from the host vehicle shall be +5.0 ±1.0 Vdc at 5 mA maximum, and shall be used as a reference in making analog-to-digital conversions. The reference voltages shall be protected against short circuits.

3.5.2 Controls. Two rotary switches shall be provided for selection of target address code. Switch positions and equivalent codes shall be as listed in TABLE II.

3.5.3 Electrical connections. The transponder shall be equipped with the connectors specified in TABLE III. Pin assignment for J3, J4 and J5 are listed in TABLE I.

TABLE III. Electrical connections.

Reference designation	Type	Function
J1	Type N female	Antenna port
J2	Type N female	Antenna port
J3	MS3110E-12-10P	Power
J4	MS3110E-18-32P	Telemetry inputs
J5	MS3110E-22-55P	Command outputs

3.6 Finish and Color. The equipment shall be semi-gloss purple NO. 27144, per FED-STD-595. Lettering shall be gloss white NO. 17886, per FED-STD-595.

3.7 Workmanship. The transponder shall be manufactured and assembled in accordance with MIL-STD-454, Requirement 9.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Ouality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified herein, all inspections shall be performed in accordance with the test conditions specified in 3.3.9 and 4.5.2. Inspections to demonstrate compliance with the requirements of MIL-E-5400 shall use the procedures of MIL-T-5422, except three cycles of the humidity tests shall be performed instead of 10.

4.4 First article inspections. First article inspections, when required (see 6.2.1), shall be conducted by the contractor on a transponder representative of the production transponders to be supplied under the contract.

4.4.1 First article inspection data. The contractor shall prepare a first article inspection report including all data collected in conducting these inspections (see 6.2.2).

4.4.2 Scope of inspection. First article inspections shall include all inspections deemed necessary by the procuring activity to determine that the transponder meets all the requirements of this specification and the contract (see 6.2.1).

4.4.3 First article approval. Approval of the first article shall be by the procuring activity upon satisfactory completion of all inspections. No production transponders shall be delivered prior to the approval of the first article. Prefabrication of production transponders prior to the approval of the first article is at the contractor's own risk. The approved

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first article shall be retained by the contractor for his use in the fabrication and testing of transponders to be submitted for acceptance. The first article shall not be considered as one of the transponders under the contract.

4.5 Quality conformance inspections. The contractor shall furnish all samples and shall be responsible for accomplishing the quality conformance inspections. All inspection and testing shall be subject to monitoring by the Government inspector. The contractor shall prepare test reports showing quantitative results for all quality conformance inspections (see 6.2.2). Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Quality conformance inspections shall consist of the following:

a. Individual tests (see 4.5.1)

b. Reliability assurance tests (see 4.5.2)

c. Special tests (see 4.5.3)

4.5.1 Individual tests. Each transponder submitted for acceptance shall be subjected to the individual tests. These tests shall demonstrate compliance with the requirements or material, operational performance and reliability. As a minimum, each transponder accepted shall have passed the following tests:

a. Examination of product (see 4.5.1.1)

b. Operational test (see 4.5.1.2)

c. Manufacturing burn-in test (see 4.5.1.3)

4.5.1.1 Examination of product. Each transponder shall be examined carefully to determine that the material and workmanship requirements have been met.

4.5.1.2 Operational test. The contractor shall conduct this test on each transponder in accordance with the performance requirements listed in TABLE IV. These tests are to be conducted under standard conditions with only verification of TTL (0 to +5 Vdc) levels required.

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TABLE IV. Operational test.

Test requirement	Requirement paragraph
1. Warm-up time	3.3.10
2. Uplink modulation characteristics	3.4.3
3. Downlink modulation characteristics	3.4.4
4. Command characteristics	3.4.5
5. Telemetry input characteristics	3.4.6
6. Power requirements	3.3.11.1
7. Detail requirements*	3.5

*NOTE: The following paragraphs are exempted from these tests: 3.5.1.1.2, 3.5.1.1.3, 3.5.1.2.2, 3.5.1.2.4, and 3.5.1.2.5.

4.5.1.3 Manufacturing burn-in test. Each transponder shall be operated under the conditions specified herein for a period of ten hours without failure. A failure shall be anything which causes malfunctioning of the equipment. Only those adjustments shall be permitted which can be made by using such controls and adjustments that are accessible to the operator during the normal use of the transponder. This test shall be deleted if the reliability assurance tests include a test on each transponder which requires at least ten hours of transponder on time. The test shall be run at ambient temperature and humidity conditions. Vibration shall be at any non-resonant frequency between 20 and 30 Hz at a level of ± 3 gs. The transponder shall be vibrated (without vibration isolators) for a period of ten minutes prior to the beginning of the ten hour period of operation. Where feasible, the transponder shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of required performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for five minutes and lateral to the plane for five minutes. Where it is not feasible to vibrate the transponder in two directions, the vertical direction shall be used. During the ten hour period of operation following the ten minute vibration period, the transponder shall be mechanically cycled periodically through its various phases of operation. Should a

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failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future transponders. A record shall be kept of all failures. The ten hour period may be composed of two five hour periods to conform with standard working hours.

4.5.2 Reliability assurance tests. Reliability assurance tests shall be conducted in accordance with MIL-STD-781. Unless otherwise specified in the contract or purchase order (see 6.2.1), tests as required by both the qualification phases and the production acceptance (sampling) phase shall be conducted. Classification of failures shall be in accordance with MIL-STD-781 and MIL-STD-2074 (see 3.3.2.3). The test details such as the length of the test cycle, the length of the heat portion of the cycle, the performance characteristics to be measured, special failure criteria, preventive maintenance to be allowed during the test and so forth, shall be part of the test procedures to be prepared prior to the beginning of the reliability assurance tests.

4.5.2.1 Qualification phase. Prior to the acceptance of transponders under the contract or purchase order, no less than three transponders shall be tested and shall have passed in accordance with test plan XIV C (Chart 15) of MIL-STD-781. For the qualification phase the test level shall be:

a. Temperature (cycling): -40 to +55°C

b. Vibration: 2.2g ±10 percent at any non-resonant frequency between 20 and 60 Hz

c. Input voltage: +27 to +29.5 Vdc

d. Transponder on-off cycling: Transponder off during cooling cycle and on during heating cycle.

4.5.2.2 All transponder production reliability acceptance test. Each transponder produced, except those submitted for the qualification phase (4.5.2.1) shall be tested for 50 failure free hours of operating time. The test level shall be the same as the qualification phase. Prior to the 50 hour failure free test, a burn-in period may be used. If the burn-in period is used, the details thereof shall be included in the procedures. To determine whether the MTBF is being met at any time during the contract, the operating test hours and the failures thereon (not counting burn-in failures or burn-in operating time) shall be totaled and the results compared with the accept-reject criteria of Figure C-13 of MIL-STD-781. These totals shall accumulate so that at any one time the experience from the beginning of the contract is included. The procuring activity shall be notified any time that the current totals of test hours and test failures plotted on

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Figure C-13 of MIL-STD-781 show a reject situation (see 6.2.2). The procuring activity reserves the right to stop the acceptance of transponders at any time that a reject situation exists pending a review of the contractor's efforts to improve the transponder, the transponder parts, the transponder workmanship, and so forth, so that the entire compilation will show other than a reject decision (see 6.2.1).

4.5.3 Special tests. When specified in the contract (see 6.2.1), special tests shall be conducted for the purpose of checking the effect of any design or material change on the performance of the transponder and to assure adequate quality control. The transponder selected for special tests may be selected from transponders previously subjected to the sampling or reliability assurance tests.

4.5.3.1 Special test schedule. Selection of transponders for special tests shall be made as follows:

a. On one of the first ten transponders after an engineering or material change.

b. Whenever failure reports or other information indicates that additional tests are required. (This will be determined by the procuring activity.)

4.5.3.2 Scope of tests. Special tests shall consist of such tests as required by the procuring activity. Test procedures previously approved shall be used where applicable. When not applicable, the contractor shall prepare a test procedure prior to conducting the tests.

4.5.4 Transponder failure. Should a failure occur during either the reliability assurance or special tests, the following action shall be taken:

a. Determine the cause of failure.

b. Determine if the failure is due to a material defect or a design defect.

c. Prepare for procuring activity approval, the proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests (see 6.2.2).

d. Where practical, include a test in the individual test to check all transponders for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

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4.6 Test procedures. The procedures for conducting first article and quality conformance inspections shall be prepared by the contractor (see 6.2.2). The right is reserved by the procuring activity to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. MIL-T-18303 shall be used for preparation of test procedures. When approved test procedures are available from previous contracts, such procedures will be provided by the procuring activity and may be used. However, the right is reserved by the procuring activity to require modification of such procedures, including additional tests, when deemed necessary (see 6.2.1).

4.7 Production transponders. Transponders supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality, be equal to the approved first article. Each transponder shall be capable of successfully passing the same tests as imposed on the first article. Evidence of noncompliance with the above shall constitute cause for rejection, and, for transponders already accepted by the Government, it shall be the obligation of the contractor to make necessary corrections.

4.8 Presubmission testing. No item, part or complete transponder, shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply with the applicable requirements.

5. PACKAGING

5.1 General. The transponder shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or purchase order (see 6.2.1) in accordance with MIL-E-17555 and MIL-STD-794. The method of preservation for level A shall be determined in accordance with the selection chart in Appendix D of MIL-STD-794. Plastic protective caps shall be provided on all connectors.

6. NOTES

6.1 Intended use. The transponder covered by this specification shall receive uplink command signals for use in the remote control of target vehicles. The transponder is one unit of the Integrated Target Control System (ITCS), AN/USW-3(V).

6.1.1 Associated equipment. The following ITCS equipment is used with the transponder:

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Nomenclature	Description
AN/USW-4(V)	Drone Control Set
AN/TSW-10(V)	Multiple Drone Control Set
AN/ASW-36	Drone Control Set
AN/MSW-10	Drone Control Set
AN/USM-417	Target Drone Test Set
AN/USM-418	Transponder Set Test Set

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.

b. Prohibition of the use of previously approved reliability program (see 3.3.2).

c. Receiver frequency (see 3.5.1.1).

d. Transmitter frequency (see 3.5.1.2).

e. Failure alarm time delay (see 3.5.1.4.11).

f. Requirement for first article inspection (see 4.4).

g. Scope of first article inspection (see 4.4.2).

h. Reliability tests, if different than specified in 4.5.2.

i. Option to stop acceptance (see 4.5.2.2).

j. Special tests, if required (see 4.5.3).

k. Test procedures if other than specified in 4.6 and additional tests, if required (see 4.6).

l. Level of preservation-packaging (see 5.1).

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the

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provisions of DAR-7-104.9 (n)(2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph No.	Data requirement title	Applicable DID No.	Option
3.2.1	Nonstandard Part Approval Requests/Proposed Additions to Approval PPSL	DI-E-7028A	
3.3.2	Plan, Reliability Program	DI-R-7079	
3.3.2.2	Repairable Items List	DI-V-7005	
3.3.7	Requests, Nomenclature, and Identification Assignment and Approval	DI-E-21336B	
4.4.1	Report, First Article (Preproduction) Test	UDI-T-21349	
4.5	Quality Inspection, Test, Demonstration and Evaluation Report	DI-R-1724	
4.5.2	Reports, Reliability Test and Demonstration	DI-R-7034	
4.5.4	Plan, Corrective Action	DI-R-7038	
4.6	Procedures, Test	UDI-T-21347	
4.6	Procedures, Reliability Tests	DI-R-7035	

(Data item descriptions related to this specification, and identified in section 6, will be approved and listed as such in DoD 5000.19L, Vol. II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3 First article. When a first article inspection is required, the item will be tested and should be a first article sample. The first article should be a sample selected from the first 10 production transponders. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, test approval of the document's first article.

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6.4 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the change.

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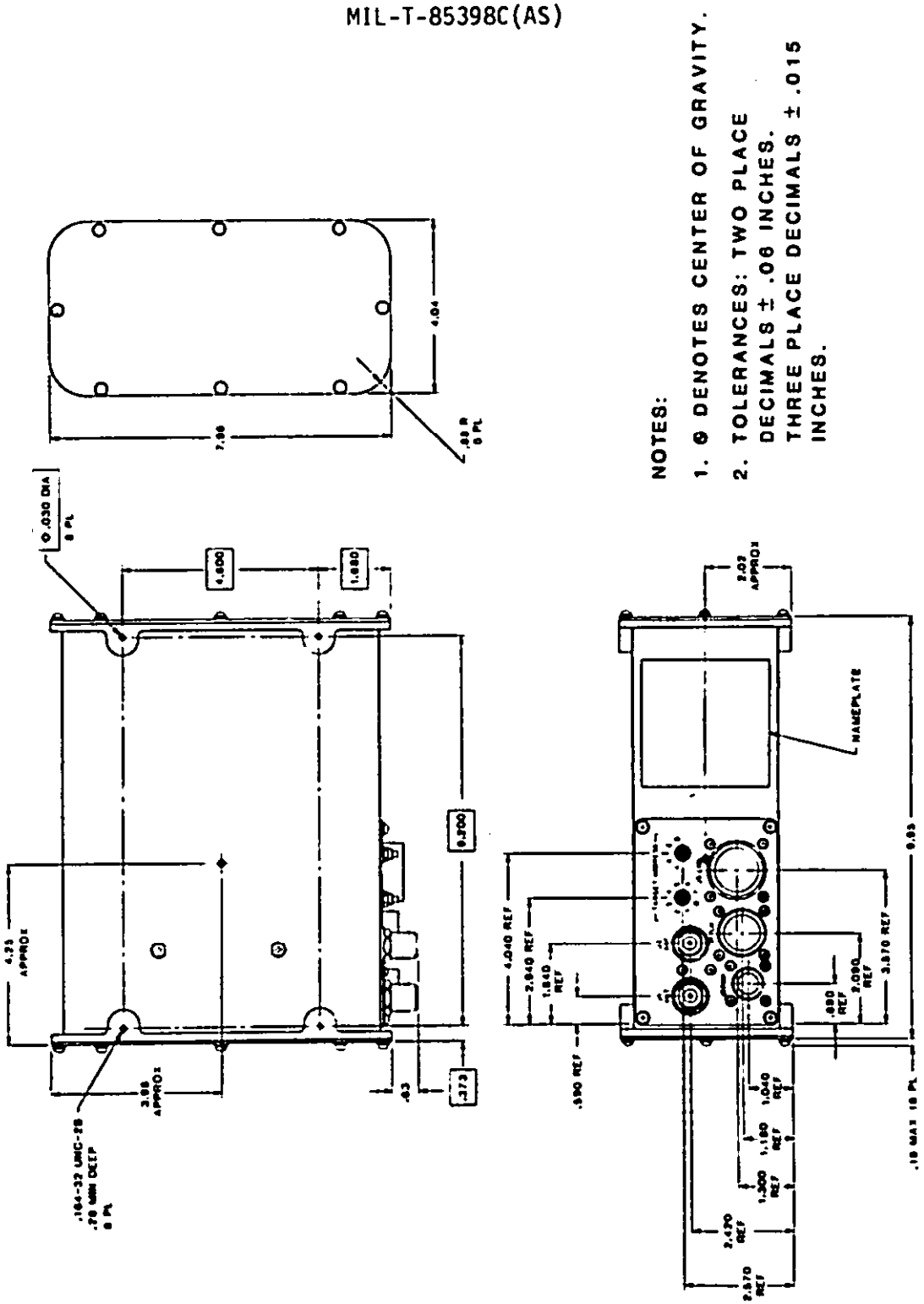
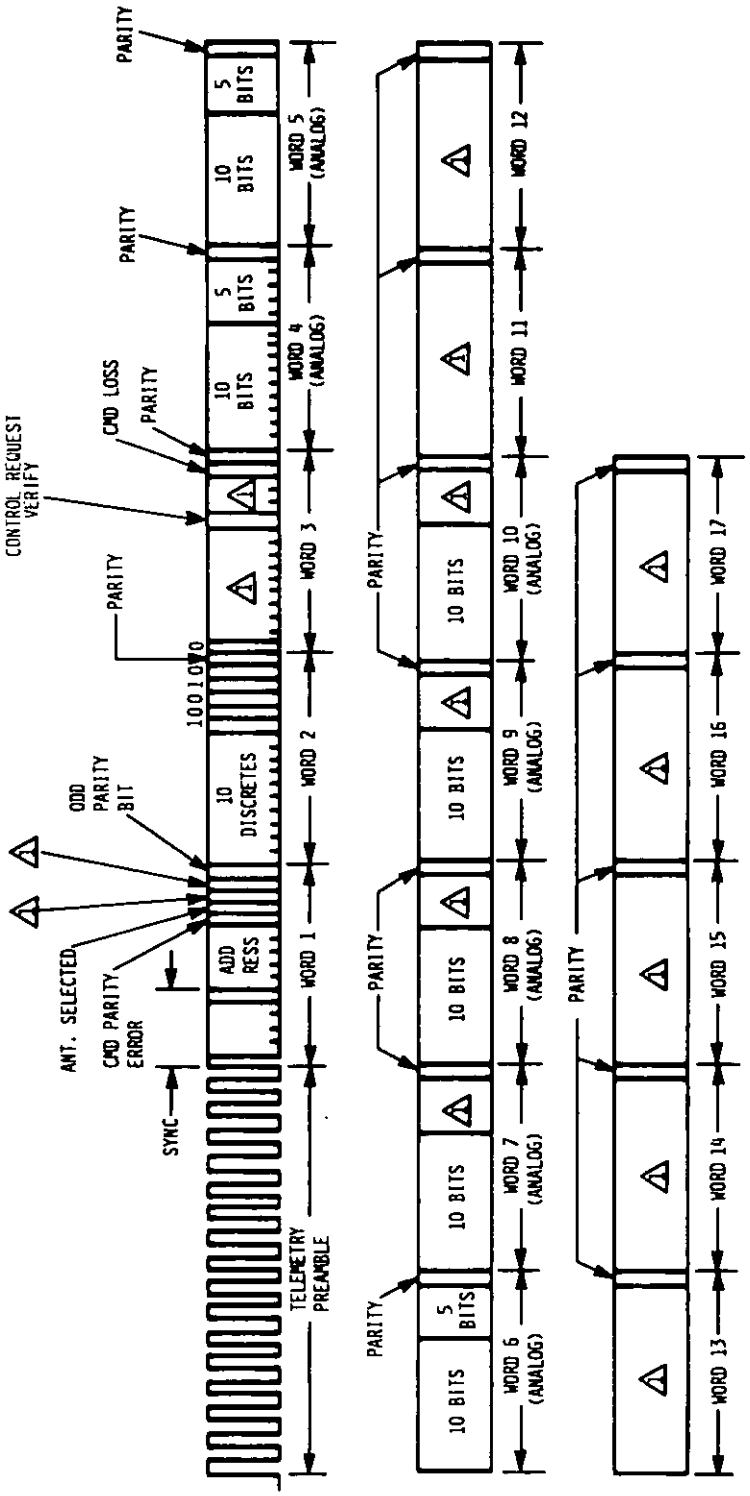


FIGURE 1. Transponder Outline

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NOTE:
1. BITS MARKED ⚠ ARE NOT ENCODED IN THE PARALLEL DATA MODE OF OPERATION IN THIS EQUIPMENT, AND ARE TRANSMITTED AS ZEROS.

FIGURE 2. Transponder downlink TM format (parallel mode).

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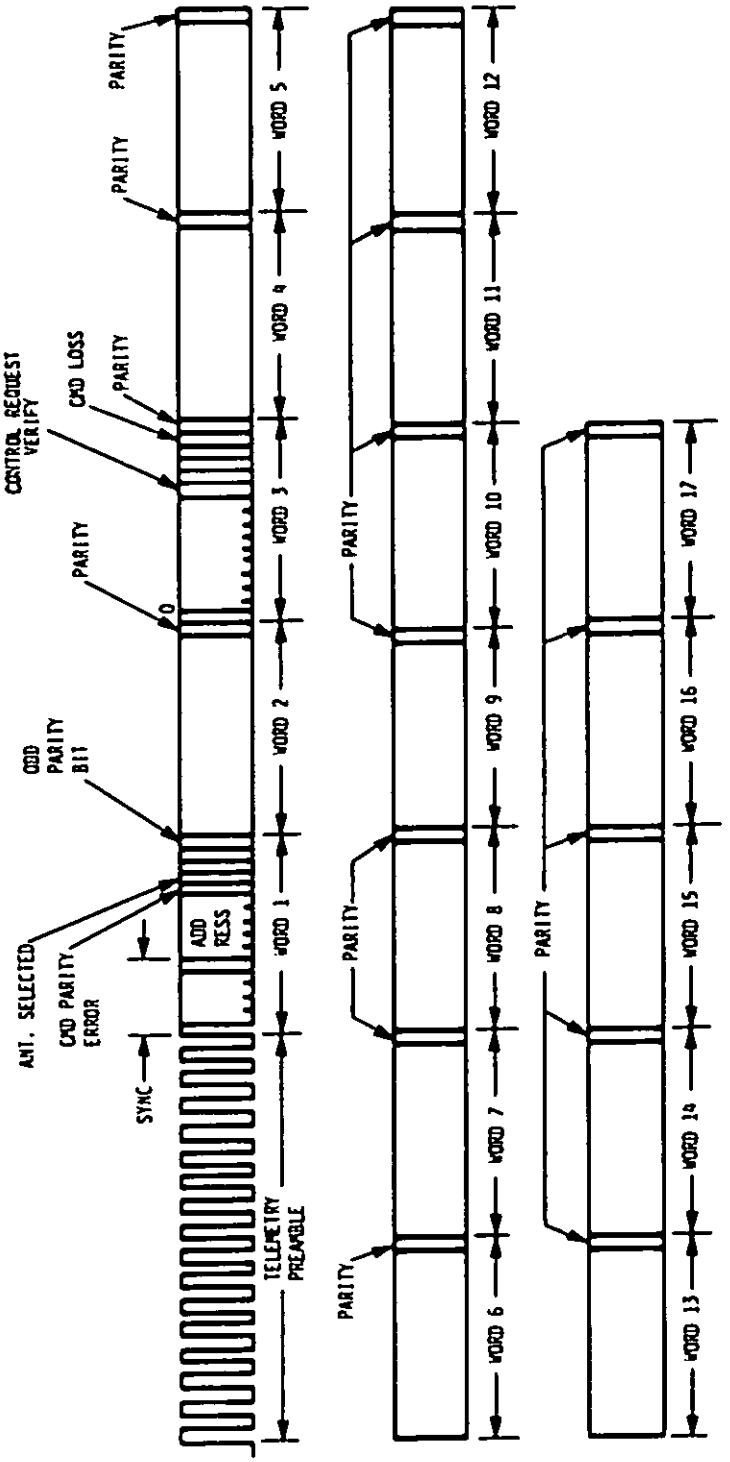


FIGURE 3. Transponder downlink TM format (serial mode).

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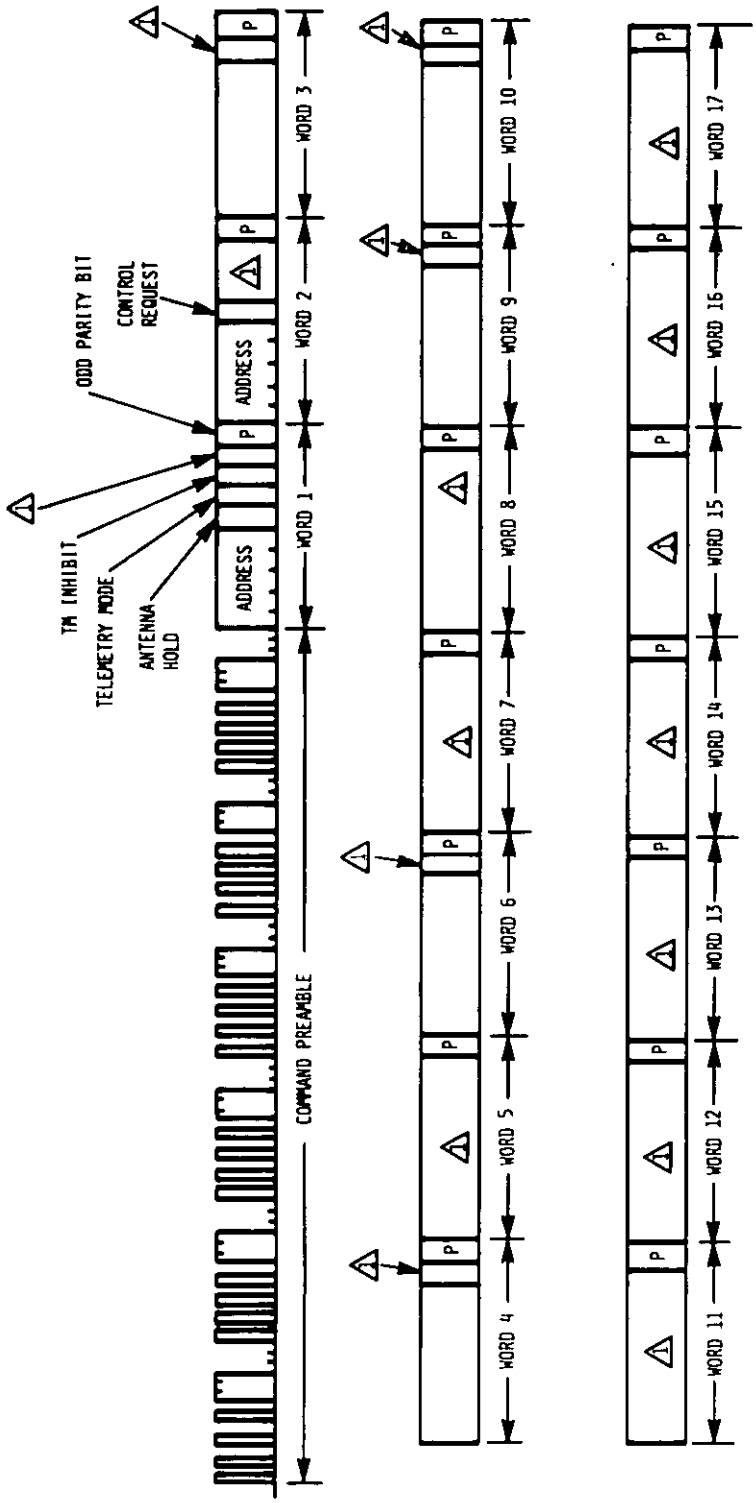


FIGURE 4. Transponder uplink command format (parallel mode).

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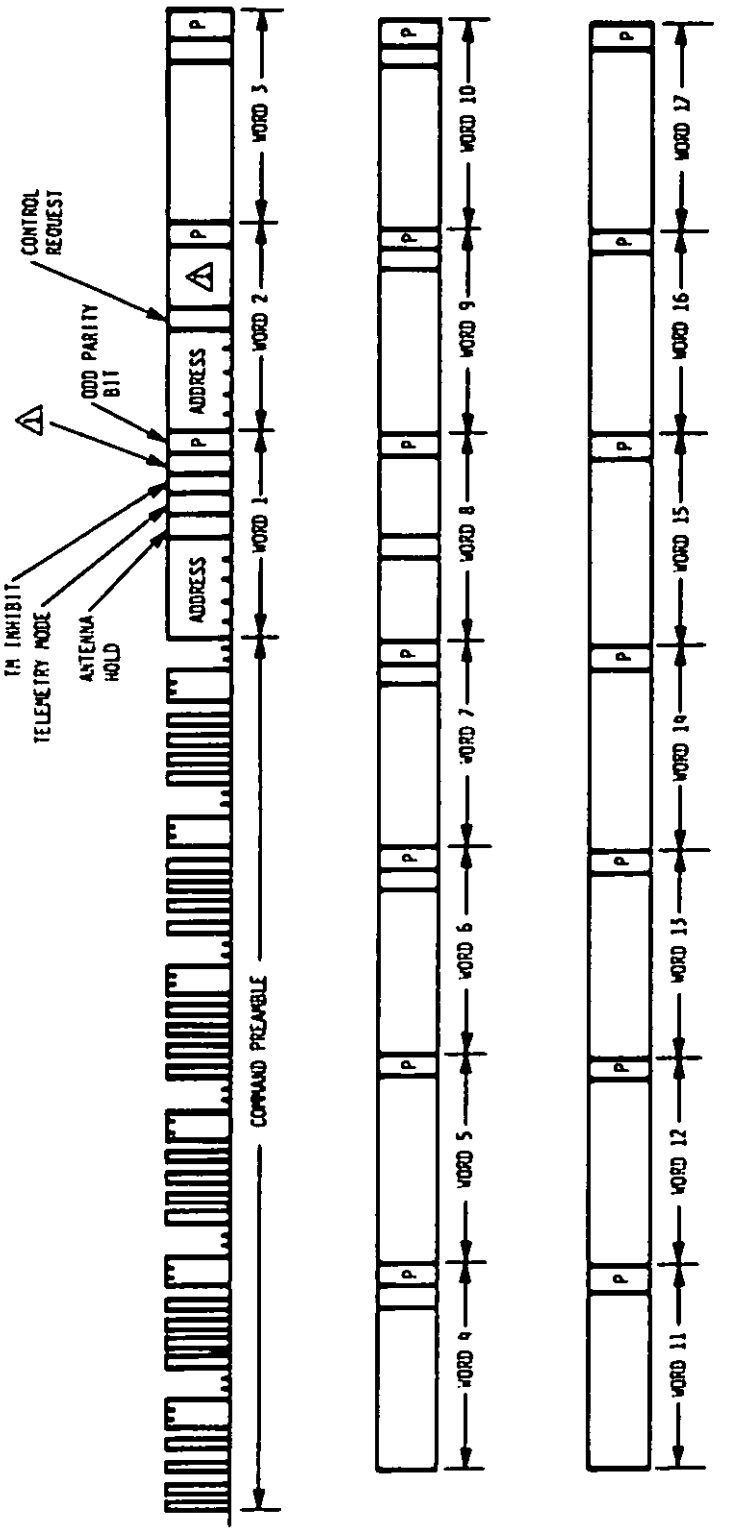


FIGURE 5. Transponder uplink command format (serial mode).

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-T-85398(AS)	TRANSPONDER SET, DRONE CONTROL AN/DKW-3A(V)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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